SAE Feature-Level Interventions for Male Skew vs Occupation Bias Decoupling

Findings Synthesis (Updated with Test-Set Results)

Author: Ayela Chughtai
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Building on: Shriya's head-level experiments, Luca's SAE analysis (Exp 05, 07, 12)

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1. Summary

Shriya showed head-level interventions cannot decouple bias reduction from gender mass suppression. I tested whether SAE feature-level interventions can break this entanglement.

Dev-set result (positive): Three SAE features at layer 11 achieve up to 34% combined bias reduction with <4% mass change and 0% PPL cost, versus L10H9 head ablation's 25% bias reduction at -20% mass. This demonstrates the entanglement is breakable at feature level in principle.

Test-set result (sobering): The two strongest features (F4077, F23772) do not generalise to held-out occupations — they are narrowly tuned to strongly female-stereotyped occupations. Only F837 (male-conditional) generalises: 11.3% bias reduction on test, 0% PPL cost, stable capability benchmarks.

Male skew result (negative): No feature at this layer reduces signed bias meaningfully. Stereotyping and male skew are mechanistically separable — stereotyping lives in sparse occupation-conditional SAE features; male skew does not.

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2. Features Discovered

Three genuine gender features at `blocks.11.hook_resid_pre` (post-L10H9 residual stream):

Feature	Role	Top tokens	Activates on	Dev abs bias red.	Test abs bias red.
F4077	Female-occ-conditional	her, herself, she	nanny, midwife, kindergarten teacher	19.9%	-5.2% (no generalisation)
F23772	Female-occ-conditional	herself, her, husband, pregnancy	nanny, midwife	11.2%	-1.5% (no generalisation)
F837	Male-occ-conditional	his, himself, he	plumber, programmer, architect	6.9%	**11.3% (generalises)**

All three confirmed genuine via logit lens (top tokens are gendered pronouns, not function words like the L0_F23406 artifact).

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3. Test-Set Evaluation (25 held-out occupations, 250 prompts)

Intervention	Abs Bias	Abs Bias Red.	Signed Bias	Gender Mass	PPL Δ	LAMBADA	BLiMP	Winogender M%	CrowS-Pairs
Baseline	0.069	—	0.069	0.160	—	15.0%	99.5%	80.5%	60.7%
F837	0.061	11.3%	0.061	0.159	0.0%	15.0%	99.5%	78.0%	60.7%
F4077 + F837	0.066	5.5%	0.065	0.158	0.0%	15.0%	99.5%	81.0%	60.7%
F4077 (half) + F837	0.064	8.3%	0.063	0.158	0.0%	15.0%	99.5%	78.5%	60.7%
All three	0.067	4.0%	0.066	0.158	0.0%	15.0%	99.5%	88.5%	60.7%

Bootstrap 95% CIs computed on all bias metrics (10,000 resamples). PPL, LAMBADA, BLiMP completely unchanged across all interventions.

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4. Key Takeaways

What worked

- **Feature-level decoupling is real on dev set.** SAE features break Shriya's head-level entanglement between bias reduction and mass suppression. This is because L10H9 contains complementary male and female sub-features — head ablation removes both (causing mass collapse), while feature clamping targets one direction only.
- **Zero capability cost.** 0.0% PPL change, stable LAMBADA, BLiMP, CrowS-Pairs across all interventions and both splits.
- **F837 generalises.** 11.3% abs bias reduction on held-out test occupations with 0% PPL cost and slight Winogender improvement (80.5% → 78.0%).

What didn't work

- **F4077 and F23772 are occupation-specific, not general.** They fire almost exclusively on a narrow cluster of highly female-stereotyped occupations (nanny, midwife, kindergarten teacher) with near-zero activation on everything else. They don't transfer to test-set occupations like pharmacist, geologist, or actuary.
- **No feature reduces male skew.** Signed bias is essentially unchanged across all interventions. Male-default preference is not encoded in sparse SAE features at this layer.
- **"All three" combined increases Winogender male preference** (80.5% → 88.5%), suggesting over-intervention on the female-conditional features pushes the model further toward male default.

Implications for the paper

1. **Stereotyping and male skew are mechanistically separable** — confirmed. Occupation-conditional stereotyping is encoded in sparse, interpretable SAE features; male skew is not.
2. **SAE features are more surgical than head ablation** — confirmed for capability preservation, but the bias reduction is narrower in scope.
3. **Generalisation is the key limitation** of SAE-based debiasing. Features may be too occupation-specific to serve as general intervention targets. This is a useful cautionary finding alongside the L0_F23406 artifact warning.

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5. Relation to Existing Experiments

Finding	Consistent with	New insight
0% PPL cost	Exp 26 (L10H9 = +1.9%)	Feature-level is strictly safer than head-level
Signed bias unchanged	Exp 26 (L10H9: +0.074 → +0.074)	Confirmed at feature level — skew lives elsewhere
F23440/F16291 dead	—	Exp 12 features not functional at this layer; likely different SAE
Generalisation gap	—	New finding: dev-set bias reduction overstates feature generality

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6. Next Steps

- Search layers 9–10 SAEs for male-skew-encoding features
- Test broader feature discovery (correlation with signed bias, not just differential activation)
- Investigate whether F837's better generalisation reflects a structural asymmetry in how male vs female stereotyping is encoded